

Task 1: Python File Handling and Automation

Objective:

To understand file handling, automation of file operations, and exception handling in Python.

Tools Used:

Python, OS Module, Shutil Module

Code:

```
import os
import shutil

def write_file():
    try:
        with open("sample.txt","w") as f:
            f.write("Hello, this is my first file handling task.")
            print("File written successfully")
    except Exception as e:
        print("Error:",e)

write_file()

def read_file():
    try:
        with open("sample.txt","r") as f:
```

```
        data= f.read()
        print("File content:")
        print(data)
    except FileNotFoundError:
        print("File not found")
read_file()

def rename_file():
    try:
        if not os.path.exists("renamed_sample.txt"):
            os.rename("sample.txt", "renamed_sample.txt")
            print("File renamed successfully")
        else:
            print("File already exists, skipping rename")
    except FileNotFoundError:
        print("File not found")
rename_file()

os.makedirs("test_folder", exist_ok=True)

def move_file():
    try:

        if not os.path.exists("test_folder"):
            os.makedirs("test_folder")
```

```
        if os.path.exists("renamed_sample.txt"):
            shutil.move("renamed_sample.txt",
"test_folder/renamed_sample.txt")
            print("File moved successfully")
        else:
            print("Source file not found")
    except Exception as e:
        print("Error:",e)
move_file()
```

```
def delete_file():
    try:
        os.remove("test_folder/renamed_sample.txt")
        print("File deleted successfully")
    except FileNotFoundError:
        print("File not found")
delete_file()
```

Output Screenshots:

- File created:

```
In [11]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task1_file_handling.py'
--wdir
File written successfully
```

- File read:

```
In [12]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task1_file_handling.py'
--wdir
File content:
Hello, this is my first file handling task.
```

- File rename, move, delete:

```
In [24]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task1_file_handling.py'
--wdir
File written successfully
File content:
Hello, this is my first file handling task.
File renamed successfully
File moved successfully
File deleted successfully
```

Short Explanation:

This program performs file operations like writing, reading, renaming, moving, and deleting files using Python. Exception handling is used to handle errors.